

READING ATTENTION AT A DEPTH THAT DOES NOT MATTER: TRANSPORT-INFORMED TOKEN ALLOCATION FOR VISION–LANGUAGE MODELS

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ABSTRACT

Vision–language models on high-resolution imagery must decide where to spend a fixed visual-token budget; evidence not given resolution is destroyed before the language model runs. The prevailing remedy is attention-guided cropping: read the model’s own attention over image positions, crop where it peaks, re-encode. Every such method must choose a depth at which to read that map, and published methods hand-pick a single layer or average a fixed block. We show that this choice dominates every other design decision, and that the depth these methods read is not the depth at which the image reaches the answer. Masking text-to-image attention layer by layer localises that transport to the first two thirds of the stack — blocking every layer from layer 16 of 28 onward flips no answers — at the same fraction of depth, 0.57, on every architecture we test, entirely before the band these methods read. Two policies follow from that measurement rather than from tuning. **Transport-Weighted Read-out** (TWR) replaces layer selection with one closed-form ridge solve over the whole depth profile, yielding a signed filter that gives *negative* weight to most of the layers a conventional read-out adds at +1. **Transport-Scheduled Resolution** (TSR) discards 90% of visual tokens at the boundary, where they are causally inert, and spends the freed compute on encoding resolution. At matched compute TWR beats every published placement rule in all nine pooled comparisons, by 7.9 and 8.9 points over the strongest, and lands its crop on the evidence 63.9% and 55.0% of the time against 46.6% and 39.3%. TSR converts a checkpoint’s resolution headroom at rate 1.01 ($r = 0.966$) across four checkpoints and three benchmarks. Against models at their *native* dynamic resolution — $5.5\times$ the budget — both are statistically indistinguishable using 14–18% of the visual-token compute, whereas published rules fall 9.9–14.2 points short of it.

1 INTRODUCTION

A vision–language model asked “what material is the glove?” about a 2250×1500 photograph must first encode the image into visual tokens. At any finite budget the glove may occupy a fraction of a token, and the evidence is gone before the language model runs. The model still *localises* it — its attention concentrates on the right region — but it cannot *perceive* it.

The standard remedy is attention-guided cropping: run the model once, read its attention over image positions, crop where the attention peaks, and re-encode that crop at higher density (Zhang et al., 2023; Shen et al., 2024). Every method in this family faces one decision that is rarely examined: **at which layer should the attention map be read?** Published methods pick a single mid-network layer or average a fixed block of late layers.

We show three things.

(i) The depth choice dominates. Cropping at the peak of one hand-picked mid layer scores 24.6% and 33.9% on two checkpoints — the four-option chance rate, and far below not cropping at all (62.3%, 57.4%). The same pipeline with a better depth read-out scores 77.2% and 71.3%.

(ii) **Everyone reads at the wrong depth, and we can say why.** Masking text tokens’ attention to the image, layer by layer, shows that image-to-text transport is *complete* by layer 16 of 28: blocking every layer from 16 onward flips no answers. Yet every method in this family — including ours — reads its map at layers 17–22. The attention these methods consume is a residue of where the model looked, not the channel carrying the evidence.

(iii) **Two allocation policies follow from that measurement**, with complementary strengths, and at native resolution they match a model spending $5.5\times$ the compute. **Transport-Weighted Read-out** (TWR, §4) stops selecting a depth and instead re-weights all of them, by a single closed-form ridge solve whose fitted weights are signed: of the layers the conventional block mean adds at $+1$, seven receive negative weight on both checkpoints, and the band’s net weight is close to zero rather than positive. **Transport-Scheduled Resolution** (TSR, §5) drops 90% of visual tokens at the boundary, where they no longer carry the image, and spends the saved compute on encoding resolution instead.

2 SETUP

Benchmarks. V*Bench (Wu & Xie, 2024) ($n = 191$; 115 single-instance, 76 cross-instance) and HR-Bench 4K/8K (Wang et al., 2024) ($n = 800$ each; 400 per stratum). Both are public; we build no datasets. Questions are four-way multiple choice, scored by the arg-max over option-letter log-probabilities, so results are deterministic.

The equal-compute bar. A two-pass method that localises at 300 tokens and answers on a crop at 300 tokens spends 600 visual tokens. We therefore compare against `uniform@600`: the same budget spent on one higher-resolution pass. Published work in this area does not budget-match; §6 shows what that hides. Where pruning changes cost mid-network we measure **token-layers** (tokens $\times$ layers traversed); the bar is $600 \times 28 = 16,800$.

Stratification. Every result is reported separately for *single-instance* questions (one object’s property) and *cross-instance* questions (a relation between two). §6.2 shows these behave oppositely and that pooling them conceals it.

Statistics. Paired item bootstrap, 8000 resamples. Learned components use GroupKFold(5) by item $\times$ 3 seeds. A claim must hold on at least two checkpoints or it is reported as rejected.

3 WHERE THE IMAGE REACHES THE ANSWER

We install a hook that masks a chosen set of query positions from attending to the image-token columns, at chosen layers only, and measure the KL divergence at the output.

Table 1: Causal masking of text-to-image attention (Qwen3-VL-2B / Qwen2-VL-7B, $n = 40$).

mask scope (all layers)	KL at output	answers flipped
the answer token ’s row only	0.025	3%
all text rows after the image	1.931	42% / 45%

Blocking the answer-emission token from seeing any image token, at every layer, changes almost nothing (Table 1). Visual evidence reaches the answer through the *text* tokens, which consolidated it upstream. Masking one layer at a time for all text rows localises that consolidation: it peaks at L11 (Qwen3) and L14–15 (Qwen2), prefix masks $L_{0..\ell}$ saturate by $\ell = 12$ –16, and suffix masks $L_{\ell..end}$ fall to ≈ 0 from $\ell = 16$ (0.014 and 0.029; zero answer flips from $\ell = 20$). On a 36-layer checkpoint the same behaviour appears at L21, i.e. the boundary transfers as $0.57\times$ depth.

The consequence. The read-out band used by every method in this family (L17–20 on Qwen3, L19–22 on Qwen2) lies *entirely after* the transport window. Those layers’ image attention is causally inert (KL ≈ 0.001). This predicts, and we confirm, three things: depth *re-weighting* helps while depth *selection* never does (§4); tokens can be dropped at the boundary for free (§5); and placement rules that all read the same inert depth separate only modestly.

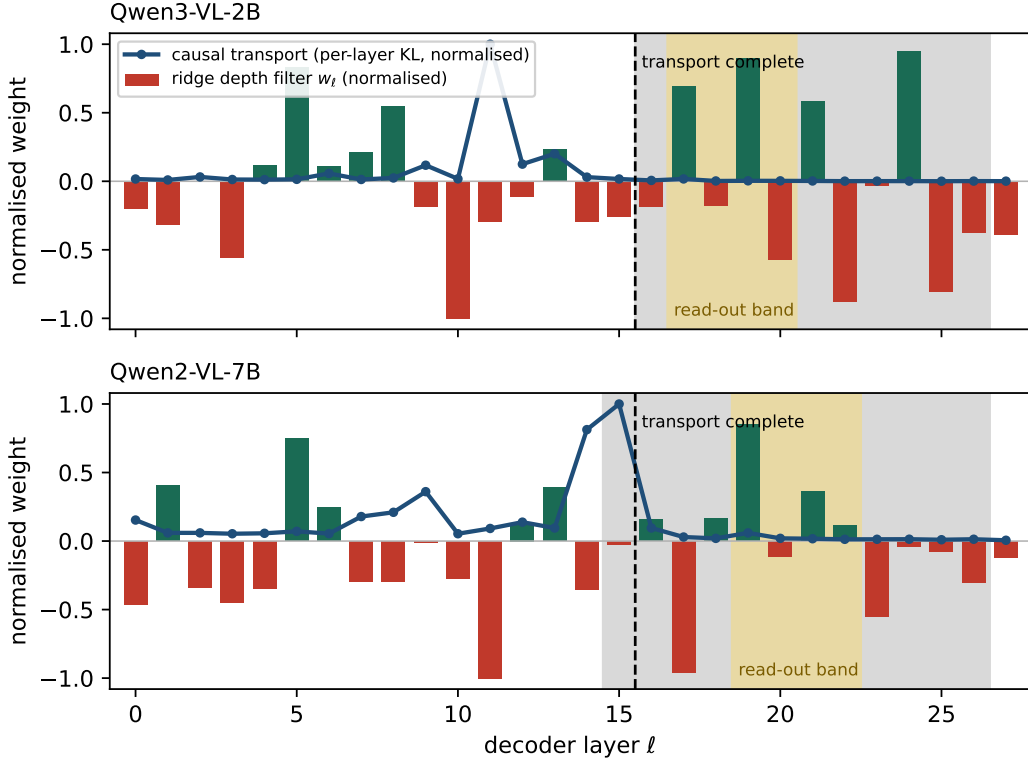


Figure 1: **The fitted depth filter sits after the transport window.** Bars: TWR’s weights w_ℓ (green positive, red negative), normalised. Line: the causal transport profile — the output KL caused by masking text-to-image attention at layer ℓ alone. Grey band: the layers the conventional block-mean read-out averages at +1. Yellow band: the question-conditioned read-out band. Dashed line: the depth beyond which masking *all* remaining layers flips no answers. The filter assigns negative weight to seven of the eleven block-mean layers here, and to seven of twelve on the other checkpoint; the band’s net weight is +0.0006 and −0.0005, i.e. approximately zero, against the +1 per layer a block mean applies. We do *not* claim the positive mass sits after the boundary: that holds on this checkpoint and reverses on the other, so by the two-model rule of §A it is not a result.

Selection stops mattering exactly at the boundary. Before it, the choice of which tokens to keep is worth +8.4 and +8.9 points over random. At and after it, ranking is worth nothing: random keep equals attention keep on both models at every keep ratio. The boundary is not merely where pruning becomes cheap — it is where selection ceases to matter.

4 POLICY I: TRANSPORT-WEIGHTED READ-OUT (TWR)

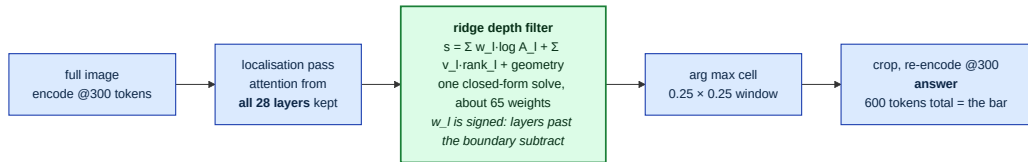


Figure 2: **Policy I: Transport-Weighted Read-out (TWR).** One localisation pass, every layer’s attention kept, a single closed-form solve that re-weights depths, and a crop. Nothing is selected: all 28 layers enter the score, and the fitted weights are signed, so layers past the transport boundary subtract rather than add.

Localise at 300 tokens with the full prompt (the read-out must be taken at the answer-emission position; without the answer instruction the map loses 36–42 points). Read the final prompt token’s attention over image cells at all 28 layers. Score each cell by a single closed-form ridge solve,

$$s(\text{cell}) = \sum_{\ell} w_{\ell} \log A_{\ell} + \sum_{\ell} v_{\ell} \text{rank}_{\ell} + a \log \text{nb} + \text{geometry},$$

fit on as few as 50 boxed examples. Crop $W=0.25$ at the arg-max, re-encode at 300 tokens, answer.

The fitted w_{ℓ} is the depth filter (Figure 1). It takes both signs: it assigns *negative* weight to 7 of the 11 layers the conventional block-mean read-out adds at +1, and its largest positive weight falls in the band where attention first becomes question-conditioned. There is no boosting, no learning rate, no early stopping, no seed averaging — given the data the solution is unique.

Why re-weighting and not selection. On the items the conventional read-out gets wrong, the filter ranks the target cell above the cell the arg-max chose on 57% and 71% of items, against 13% and 12% for the block mean. No monotone summary of the depth profile reproduces this — not magnitude, neighbourhood, isolation, or a windowed sum — because the correction requires subtraction, and a mean can only add.

Ablations. A gradient-boosted tree over the same features is statistically indistinguishable from the ridge solve on all four architectures, so the closed form costs nothing. Capacity is not the limit at this sample size: a spatial CNN is *significantly worse* on the 7B (−8.9 and −9.4 at two window sizes), and nine random seeds, flip augmentation and ensembling all land inside the noise floor. Roughly 50 boxed examples recover 70% of the full-data gain. A training-free variant — the maximum over the four question-conditioned layers — reaches 68.9 and 60.7, above every published rule but 3.7 and 9.5 points below the fitted filter.

5 POLICY II: TRANSPORT-SCHEDULED RESOLUTION (TSR)

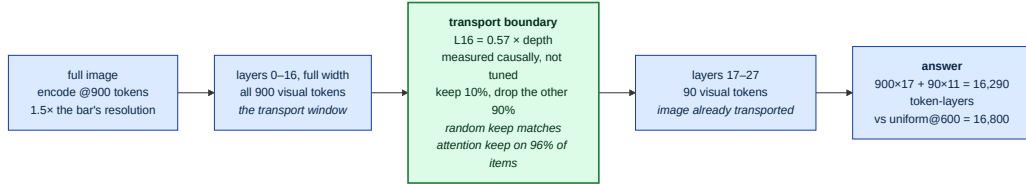


Figure 3: **Policy II, transport-scheduled resolution.** Spend the compute the inert late-layer visual tokens would have used on resolution inside the window where the image is actually read. The prune depth is read off the causal measurement rather than tuned, and nothing spatial is discarded before the boundary.

Because visual tokens are inert past the boundary, discard them there and spend the saving on pixels: encode at 900 tokens, run full width through L0–L16, keep the top 10% at L16, run L17–27 narrow. Token-layers $900 \times 17 + 90 \times 11 = 16,290$ — 96% of the bar, for $1.5 \times$ the resolution. No crop.

Pruning is free: 0.0–0.5 points in 9 of 10 measured cells, at every keep ratio (10/25/50%), with any ranking including random. Output probabilities change on 99% of items while the arg-max changes on 2% — the tokens are inert, not absent.

An identity. Across 4 checkpoints $\times$ 3 benchmarks, TSR’s gain equals the checkpoint’s *measurable resolution headroom* $\text{acc}(900) - \text{acc}(600)$ on that stratum: $r = 0.966$, slope 1.01, mean absolute error 0.7 points over 10 stratum-cells. The headroom needs two baseline runs and no method at all, so TSR’s benefit is *predictable before deployment* — and it explains the cells where TSR gains little: those are the cells where the checkpoint gains little from resolution in the first place, including one where the baseline itself degrades from 600 to 900 tokens.

The boundary is the right depth, not merely a cheap one. Pruning the same fraction at layer 2 (the FastV default) costs 4.7 points more; a causally-scheduled taper that prunes 50% inside the

transport window costs 5.8 points more than one cut at the boundary and loses to PyramidDrop’s fixed schedule; and pruning at 0.14 of depth rather than 0.57 on the 36-layer checkpoint costs 7.9 points pooled. Conversely, pruning *after* the boundary saves less for no additional accuracy. One cut, at the measured boundary, dominates every schedule we tried.

6 RESULTS

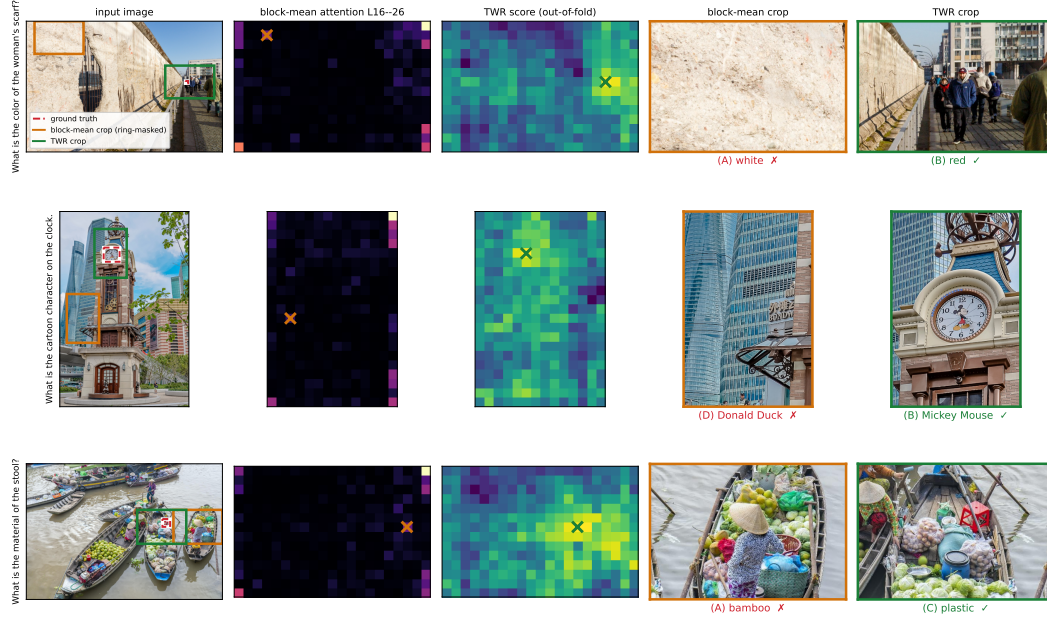


Figure 4: **What the two read-out rules see, on real V*Bench items (Qwen3-VL-2B).** The block-mean arg-max is shown as the paper evaluates it, which masks the grid’s outer ring before taking the arg-max — and that mask is doing very unequal amounts of work. Without it, the block-mean arg-max lands in the grid’s *last column*, the position before a raster row-wrap, on 84.8% and 61.8% of items across the two checkpoints against a 5.0% chance rate, and covers half the ground-truth box on only 13.6% of items; the ring mask lifts that to 46.6% and 39.3% across the two checkpoints. TWR covers 63.9% and 55.0%, and barely uses the mask: it is worth +0.012 and +0.003 mean coverage to TWR against +0.336 and +0.079 to block-mean, an order of magnitude on both. TWR removes the serialisation sink by giving the post-boundary layers negative weight rather than by patching the geometry. *Examples are not hand-picked*: we take single-instance items where the TWR crop covers the box and the block-mean crop does not *and* the answers flip accordingly, restricted to items where the recomputed out-of-fold TWR cell reproduces the logged run exactly. Nine items qualify; these are the first three by question id.

TWR beats every published rule on every cell where that rule was run — 9 of 9 paired contrasts, all CIs clear: +3.8 to +15.9 against ViCrop-style, +7.9 and +8.9 against LASER, +31.9 and +34.6 against fixed-layer. At matched compute the published rules are worth −1.6 to +3.1 against simply spending the tokens on resolution.

6.1 NATIVE DYNAMIC RESOLUTION

The field publishes these benchmarks at the model’s native dynamic resolution. On V*Bench that is a median of 3,290 visual tokens — $5.5\times$ our bar — and 76.3% accuracy.

Three readings, both checkpoints. (i) TWR is *statistically indistinguishable* from native dynamic resolution (−3.7 and −1.0, CIs spanning zero) while using 18% and 14% of native’s visual-token compute — whereas every published rule is *significantly worse* than native (−9.9 to −14.2). The separation between our read-out and theirs is therefore not an artefact of the imposed budget; it is larger in the regime the field actually publishes in. (ii) Per stratum on Qwen3, TWR at 600 tokens

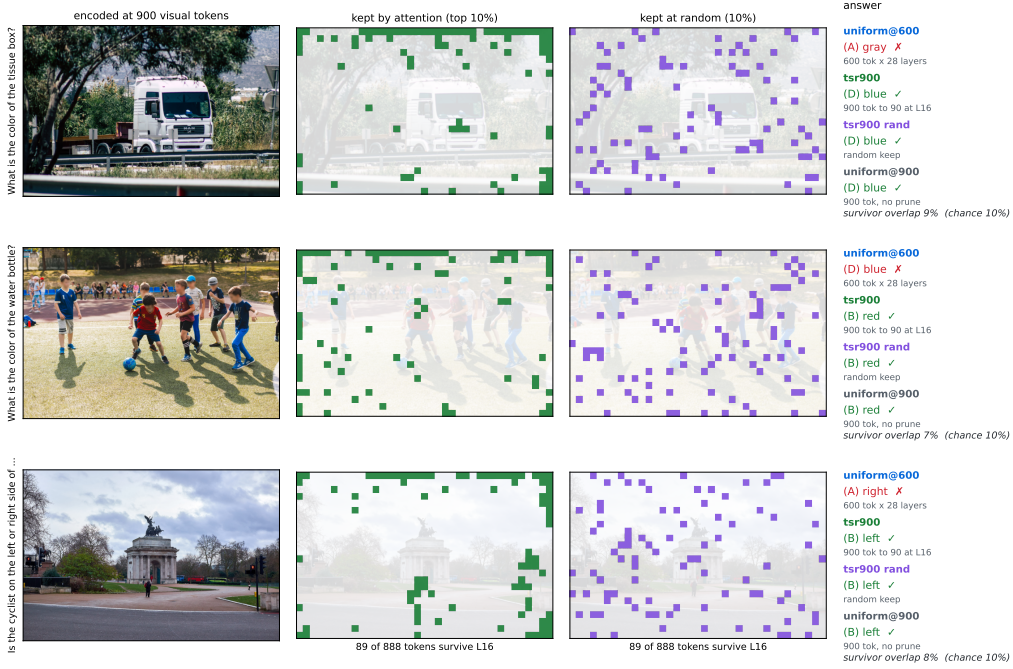


Figure 5: **What survives the transport boundary, and why the choice of survivor does not matter** (Qwen3-VL-2B). Of 888 visual tokens, 89 are carried past L16. The target encode is 900 tokens; the processor’s own grid yields 888 on these images, inside the $\pm 8\%$ tolerance of Appendix A, and the arithmetic quoted elsewhere ($900 \times 17 + 90 \times 11$) is at the target. Ranking them by attention selects mostly border tokens — the same sink Figure 4 exposes — while a random keep of the same size overlaps it by only 7–9%, at the 10% chance rate. Both give the same answer, and so does not pruning at all, on all three items; across the benchmark attention-keep and random-keep agree on 96.3% of items (70.2% vs 68.6% accuracy). The gain over the bar is resolution, not selection. Selection rule for the examples: TSR correct, the equal-compute bar wrong, and random keep agreeing with attention keep; 15 items qualify.

Table 2: All placement rules in one run, V*Bench, every arm at ~ 600 visual tokens. $\checkmark$ = 95% CI clear. $n=190$ on Qwen3 and $n=191$ on Qwen2: one item lacks the LASER arm on Qwen3 and is dropped from every arm so all rules are compared on identical items. Table 3 uses the same item sets.

method	Qwen3-VL-2B			Qwen2-VL-7B		
	single	cross	ALL	single	cross	ALL
uniform@600 (bar)	62.3	65.8	63.7	57.4	59.2	58.1
fixed layer L14	24.6	57.9	37.9	33.9	44.7	38.2
block-mean arg-max (ViCrop-style)	64.0	59.2	62.1	67.0	47.4	59.2
LASER (Zhu et al., 2026)	64.0	65.8	64.7	65.2	55.3	61.3
gate-max (ours, no training)	72.8	63.2	68.9	63.5	56.6	60.7
TWR (ours)	77.2	65.8	72.6	71.3	68.4	70.2
<i>oracle crop (upper bound)</i>	96.5	75.0	87.9	93.9	77.6	87.4

matches native on single-instance (+2.6) and TSR matches it on cross-instance (−2.6), each at a fifth of the compute. (iii) A perfectly placed 600-token crop *beats* native by +11.6 and +16.2 pooled, and by +21.7 on single-instance (Qwen3): **placement is worth more than 5–7× the compute**. This is the strongest single argument for allocation research, and it is a ceiling nobody has approached.

Table 3: **Against native dynamic resolution.** V*Bench. Our arms use 600 visual tokens; native uses 3,290 (Qwen3) and 4,320 (Qwen2), i.e. $5.5\times$ and $7.2\times$ more. ✓/✗ mark CIs clear of zero.

arm	Qwen3-VL-2B ($n=190$)		Qwen2-VL-7B ($n=191$)	
	accuracy	vs. NATIVE	accuracy	vs. NATIVE
NATIVE dynamic resolution	76.3	—	71.2	—
uniform@600 (bar)	63.7	−12.6 ✗	58.1	−13.1 ✗
block-mean arg-max	62.1	−14.2 ✗	59.2	−12.0 ✗
LASER (Zhu et al., 2026)	64.7	−11.6 ✗	61.3	−9.9 ✗
TSR (ours)	70.0	−6.3	61.8	−9.4 ✗
TWR (ours)	72.6	−3.7	70.2	−1.0
<i>oracle crop @600</i>	87.9	+11.6 ✓	87.4	+16.2 ✓

Table 4: **The scope law.** Accuracy difference vs. the equal-compute bar, per stratum, all seven benchmark $\times$ checkpoint cells. Cropping helps single-instance everywhere and is non-positive on cross-instance almost everywhere; TSR is positive on both strata in 6 of 7 cells but gives up single-instance gain.

cell	TWR (2 passes, ~ 50 boxes)		TSR (1 pass, no supervision)	
	single	cross	single	cross
V*Bench Qwen3-VL-2B	+14.8 ✓	+0.0	+3.5	+10.5 ✓
V*Bench Qwen2-VL-7B	+13.9 ✓	+9.2	+3.5	+1.3
V*Bench Qwen2.5-VL-7B	+12.2 ✓	−2.6	+12.2 ✓	+7.9
V*Bench Qwen3-VL-8B	+6.1	−3.9	+7.0	+7.9 ✓
HR-4K Qwen3-VL-2B	+10.2 ✓	−8.8 ✗	+4.0 ✓	−2.8
HR-4K Qwen2-VL-7B	+9.2 ✓	−6.0 ✗	+2.8	+0.8
HR-8K Qwen3-VL-2B	+8.8 ✓	−8.8 ✗	+6.5 ✓	+1.8
positive cells	7/7	1/7	7/7	6/7
significant losses	0	3	0	0

6.2 A SCOPE LAW

Across 14 cells (3 benchmarks $\times$ 4 checkpoints $\times$ 2 strata), attention-guided cropping helps single-instance perception in 7/7 and is non-positive on cross-instance in 6/7, significantly negative in three (−6.0, −8.8, −8.8). This holds for *every* method in the family, ours included: a tight crop magnifies one object and discards the other. TSR shows the opposite profile — positive on both strata in 6/7 cells with no significant loss anywhere — but gives up 3–10 points of single-instance gain.

The ceiling explains why cropping cannot be repaired here. With *perfect* placement, cross-instance still fails 25% and 22% of the time against 3.5% and 6.1% for single-instance. Cross-instance error is not predominantly perceptual, and no crop, multi-crop, composition or pruning schedule reached it (§8).

7 RELATED WORK

Attention-guided cropping. Liu et al. (2026) document the gap this family exploits, that a model’s attention can concentrate correctly while its answer is wrong. ViCrop (Zhang et al., 2023) crops at the peak of a fixed mid-layer attention map; ZoomEye (Shen et al., 2024) searches a tree of crops with repeated model queries. VisLens (He et al., 2026) reaches the same place without attention at all, decoding visual tokens through a tuned logit lens, matching them against the query, and feeding the crop back alongside the original image in one pass. Every attention-based member fixes the read-out depth by hand, and none of them budget-match against simply spending the same tokens on resolution — Table 2 shows that at equal compute the family is worth −1.6 to +3.1 points.

Layer-adaptive read-outs. LASER (Zhu et al., 2026) selects the attention layer per sample by contrasting the map with and without the query, and reports that no single “magic layer” suffices

— a conclusion we reach independently and extend causally. ILVAD (Xie et al., 2026) accumulates newly-activated tokens across adjacent layers; TRACE (Ye et al., 2026) identifies a three-stage depth pattern and reshapes it with trained modules; ViRGo (Tran et al., 2026) reports our cross-instance finding (“patch-based zooming . . . can split large objects and destroy global spatial context”) and resolves it with a three-way router keyed on object scale. We treat these as concurrent work reaching compatible conclusions. Our contribution relative to them is (i) the causal localisation of the transport window rather than a correlational choice of layer, (ii) a signed *re-weighting* across all depths rather than a selection among them, and (iii) budget-matched, stratified evaluation including against native resolution.

Depth selection in language models. DoLa (Chuang et al., 2024) contrasts a mature and a premature layer; SLED (Zhang et al., 2024) weights all early layers by gradient alignment; confident-layer decoding stops at an entropy valley near the output. We port each to this setting and all fail (§8) for one measurable reason: in these vision–language models final-layer entropy rises on 54–88% of items — far above the 16% reported for language models — yet per-layer accuracy still climbs monotonically to the last layer. There is no better layer to select, which is why re-weighting works where selection does not.

Visual-token pruning. FastV (Chen et al., 2024) prunes at layer 2; PyramidDrop (Xing et al., 2024) drops tokens in stages with a fixed ratio; HiRED (Arif et al., 2025) allocates a budget before the language model. These are evaluated as retention-at-reduced-cost. We derive the pruning depth from the transport measurement rather than tuning it, and show that at that depth the *choice* of surviving tokens is irrelevant — random keep matches attention keep — while one layer earlier it is worth +8.4 and +8.9 points.

8 WHAT DOES NOT WORK

Ten designs failed to make cropping serve cross-instance questions, and two more failed to combine the two policies into a single pipeline. We report them because the diagnoses constrain the design space more than the successes do. Each failed for a measured reason: per-item window sizing; peak-spanning and mass-containment windows; joint (cell, window) re-ranking under a coverage objective (a metric artefact — coverage is monotone in window size); expected-accuracy window selection; a downscaled scene composed with the crop, which dilutes it by 8.7 and 5.2 points *even at oracle placement*; multi-crop at matched budget, where the second crop genuinely converts (+7.8 ✓) but halving each crop costs −11.3; and the same at full resolution funded by pruning. Depth *selection* rules ported from the LLM literature — DoLa, SLED, confident-layer decoding, attention-JSD, rank-stability — all fail, for one reason: in these models final-layer entropy rises on 54–88% of items while per-layer accuracy still climbs monotonically to the last layer. There is no better layer to select.

9 PRACTICAL GUIDANCE

The two policies have different cost and risk profiles, and the choice does not require a question classifier. The TWR needs ~50 boxed examples, two forward passes and $1.4\text{--}2.0\times$ wall-clock; it has the higher ceiling (+14.8 on single-instance) and a 6–9 point cross-instance failure mode. TSR needs no supervision, one forward pass and $\sim 1\times$ wall-clock; it is positive on both strata in 6 of 7 cells with no significant loss anywhere, and its benefit is predictable from two baseline runs. Head to head the two are even pooled (mean difference +0.2 points over seven cells). We recommend TSR as the default and TWR where boxed supervision exists and single-instance perception dominates. We note that TWR’s pooled margin over TSR is concentrated on the two cells on which it was developed.

10 LIMITATIONS

$n = 191$ on V*Bench gives per-stratum CI half-widths near 10 points; no method is significantly positive on both strata in any single cell, which is a power limitation as much as a method one. TWR needs ~50 boxed examples and two forward passes ($1.4\text{--}2.0\times$ wall-clock); TSR needs neither. Our two core checkpoints share a model family. TWR’s pooled margin over TSR is concentrated on

the two cells it was developed on. Pruning at the boundary is free in 9 of 10 cells; the exception (HR-Bench 8K, cross-instance, $-1.8 \times$) is real.

11 CONCLUSION

Attention-guided cropping reads the model’s attention at a depth where that attention no longer carries the image to the answer. Measuring where it does — a causal masking experiment, not a heuristic — yields both a better read-out (TWR) and a free pruning point (TSR), and the two together match a model running at $5.5 \times$ the visual-token budget. The remaining gap is placement: perfect placement at 600 tokens exceeds native dynamic resolution by 11.6 points.

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A EXPERIMENTAL PROTOCOL

Models. Qwen3-VL-2B-Instruct and Qwen2-VL-7B-Instruct (core pair), Qwen2.5-VL-7B-Instruct, Qwen3-VL-8B-Instruct (36 layers), LLaVA-NeXT-7B and LLaVA-OneVision-7B (attention analyses only). `bfloat16`; eager attention wherever hooks or a prune bias are required, SDPA otherwise. We note a ~ 2 point kernel-dependent difference on the 7B between the two implementations and always compare arms computed under the same kernel.

Budgets. Images are resized so that the processor’s own `image_grid_thw` yields the target token count ($\pm 8\%$); realised counts are recorded per arm and a contrast is voided if budgets drift by more than 10%. Where pruning changes cost mid-network we report token-layers.

Read-out. The attention map is the final prompt token’s attention over image-token columns, averaged over heads *before* normalising (the order matters: normalising per head first up-weights sink-heavy heads and shifts the block-mean read-out by 11 points on the 7B). The prompt must end at the answer-emission point; a prompt ending elsewhere costs 36–42 points on the arg-max read-out.

Statistics. Paired item bootstrap with 8000 resamples. Learned components use GroupKFold(5) grouped by item $\times 3$ seeds; the headline TWR result is additionally reproduced under a disjoint fold-seed set (difference within ± 1 point). Run-to-run floor on coverage is 1–2.5 points.

Pipeline discipline. An exactly $+0.0$ contrast with a $[+0.0, +0.0]$ interval is treated as a pipeline fault, never a result; this rule caught a monkey-patch that silently failed to reach one model family. Every pruning run asserts at runtime that the loaded model’s attention module is the one that was patched.

B INTERMEDIATE METRICS SYSTEMATICALLY OVER-REPORT

Five times in this work an intermediate metric improved while end-task accuracy did not: a read-out that closed the coverage gap but converted 3.1 points worse; a ridge variant that tied on coverage and differed on the chosen cell for two-thirds of items; an early-exit localiser free in coverage but -3.7 pooled at the end task; a higher-resolution localiser that gained $+4.2$ coverage and -3.7 accuracy; and an objectness probe that retains the evidence box far better than attention (87% vs. 72%) and converts to -1.6 . We therefore report coverage only as a diagnostic and gate every claim on end-task accuracy at matched compute.

C THE TRANSPORT MEASUREMENT IN FULL

A forward pre-hook on every decoder layer adds -10^4 to the attention logits of chosen query rows over the image-token columns, at chosen layers only. Masking *all* text rows at *all* layers destroys the answer (KL 1.266 / 0.532, 42% / 45% of answers flip, $n=40$); masking only the answer token’s row at all layers does not (KL 0.025, 3% flip). Table 5 localises the window.

Single-layer masks peak at L11 on Qwen3 (KL 0.504; next highest L13 at 0.101) and at L15/L14 on Qwen2 (0.118/0.096). From L20 onward every layer is at or below 0.0023 on both models. On Qwen3-VL-8B, which has 36 layers, the same behaviour appears at L21, i.e. the same 0.57 fraction of depth.

Three independent readings of the same boundary. The masks are one. Pruning is the second: discarding 90% of visual tokens at the boundary costs $+0.0$ to $+0.5$ points across ten stratum-cells, whereas moving the same cut to $\ell=4$ costs 5.2 more on one model (0.5 on the other), and a taper that removes half the tokens at $\ell=9$ –12 loses 3.5 points against not pruning at matched tokens while buying nothing over a single cut (-1.0). Read-out depth is the third: restricting the localiser to layers $\leq L20$ costs -0.5 and $+1.6$ points of coverage, while restricting it to $\leq L16$ costs 9.4 and 22.0. Transport finishes at the boundary; the signal the ranking reads is produced after it.

Table 5: Prefix and suffix masks, all text rows, $n = 40$ per model, V*Bench at 300 tokens. The prefix mask $L_{0..l}$ saturates at the full-mask value once l passes the window; the suffix mask $L_{l..end}$ falls to zero once l is past it. Both read the boundary at the same place.

	$l=4$	$l=8$	$l=12$	$l=16$	$l=20$	$l=24$
Qwen3 prefix $L_{0..l}$ (KL)	0.082	0.549	1.278	1.259	1.215	1.257
Qwen3 suffix $L_{l..end}$ (KL)	0.967	0.880	0.320	0.014	0.002	0.001
Qwen3 suffix (answers flipped)	53%	50%	20%	12%	0%	0%
Qwen2 prefix $L_{0..l}$ (KL)	0.037	0.105	0.364	0.547	0.534	0.538
Qwen2 suffix $L_{l..end}$ (KL)	0.518	0.465	0.429	0.029	0.004	0.002
Qwen2 suffix (answers flipped)	53%	47%	40%	5%	3%	0%

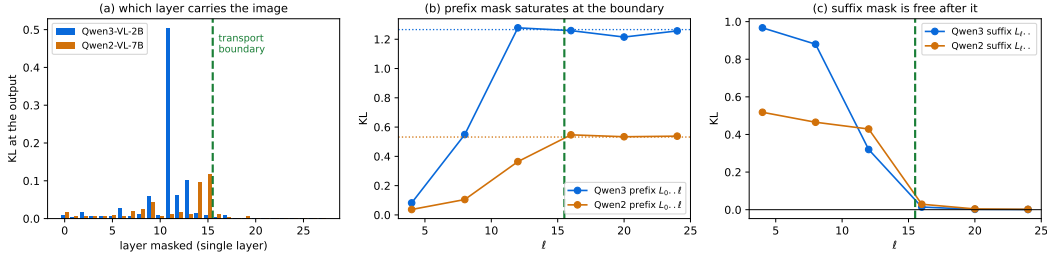


Figure 6: **The transport measurement.** (a) Masking one layer’s image attention at a time: the effect is concentrated well before the boundary and is at or below 0.0023 from L20 on both models. (b) Masking a prefix $L_{0..l}$ reaches the full-mask value (dotted) once l passes the window. (c) Masking a suffix $L_{l..end}$ costs nothing once l is past it. Three readings, one boundary.

D DEPTH STRUCTURE ACROSS CHECKPOINTS

Table 6: Where each quantity falls, per checkpoint. The transport boundary and the read-out band are fractions of depth, not fixed layer indices, which is what lets both policies transfer to a 36-layer model unchanged.

checkpoint	layers	block-mean band	read-out band	boundary	boundary / depth
Qwen3-VL-2B	28	L16–L26	L17–L20	L16	0.57
Qwen2-VL-7B	28	L15–L26	L19–L22	L16	0.57
Qwen2.5-VL-7B	28	L16–L26	—	L16	0.57
Qwen3-VL-8B	36	L20–L33	—	L21	0.57

E FULL PAIRED COMPARISON AGAINST PUBLISHED RULES

Taking these six together with the three pooled block-mean contrasts in Table 8, TWR beats a published rule in nine of nine pooled comparisons. It does not beat the equal-compute bar everywhere, which is a separate claim and the subject of the scope law. The fold-seed control matters most. Refitting TWR under three disjoint fold seeds moves the pooled result by 1.0 point or less on both models, so the margin is not an artefact of which items landed in which fold. The oracle row is the remaining headroom: even the best rule we have is 15–17 points below perfect placement.

The pooled rows against the bar are the scope law in its plainest form. Cropping wins 9–10 points where the question is about one instance, loses 6–9 where it is about a relation between two, and the two cancel to nothing in aggregate. A paper that reported only the pooled column would conclude the method does not work; a paper that reported only the single-instance column would conclude it works everywhere. Neither is true.

Table 7: TWR minus each alternative on V*Bench, paired on identical items, pooled over strata, $n = 191$ (190 for LASER, one item missing). All six published-rule contrasts clear zero on both models. Bootstrap 95% CIs, 8,000 resamples, matching the protocol in Appendix A.

contrast	Qwen3-VL-2B	Qwen2-VL-7B
– uniform@600 (equal-compute bar)	+8.9 [+0.5, +17.3] ✓	+12.0 [+4.7, +19.4] ✓
– fixed-layer crop (L14)	+34.6 [+25.7, +43.5] ✓	+31.9 [+24.1, +39.8] ✓
– block-mean arg-max	+10.5 [+4.2, +16.8] ✓	+11.0 [+4.7, +17.8] ✓
– LASER	+7.9 [+1.1, +14.7] ✓	+8.9 [+3.1, +14.7] ✓
<i>controls, not baselines</i>		
– gate-max	+3.7 [−2.1, +9.4]	+9.4 [+3.7, +15.2] ✓
– gradient-boosted tree	−0.5 [−4.7, +4.2]	+7.3 [+2.1, +13.1] ✓
– TWR refit, disjoint fold seeds	−1.0 [−3.1, +1.0]	+1.0 [−1.0, +3.1]
– oracle placement	−15.2 [−22.5, −8.4] ✗	−17.3 [−24.6, −9.9] ✗

Table 8: HR-Bench, TWR minus each alternative, $n = 400$ per stratum, 800 per cell. TWR is transferred: its weights were fit on V*Bench and applied here with nothing refit. Placement quality replicates; the scope law replicates with it.

		HR-4K Qwen3-2B	HR-4K Qwen2-7B	HR-8K Qwen3-2B
vs. bar	single	+10.2 [+5.5, +15.0] ✓	+9.2 [+4.5, +14.2] ✓	+8.8 [+3.5, +13.8] ✓
	cross	−8.8 [−14.0, −3.5] ✗	−6.0 [−11.5, −0.8] ✗	−8.8 [−14.5, −2.8] ✗
	pooled	+0.8 [−2.8, +4.4]	+1.6 [−1.9, +5.2]	+0.0 [−3.9, +4.0]
vs. block-mean	single	+7.8 [+4.0, +11.5] ✓	+15.0 [+10.2, +19.8] ✓	+5.2 [+1.2, +9.2] ✓
	cross	+1.8 [−2.8, +6.2]	+16.8 [+12.0, +21.5] ✓	+2.2 [−2.5, +6.8]
	pooled	+4.8 [+1.9, +7.6] ✓	+15.9 [+12.6, +19.2] ✓	+3.8 [+0.6, +6.9] ✓
vs. random cell	single	+32.8 [+26.5, +39.0] ✓	+24.8 [+18.8, +30.5] ✓	+24.2 [+18.0, +30.2] ✓
	cross	+10.8 [+5.0, +16.5] ✓	+15.8 [+10.5, +21.0] ✓	+7.0 [+2.0, +12.0] ✓
	pooled	+21.8 [+17.5, +26.1] ✓	+20.2 [+16.4, +24.2] ✓	+15.6 [+11.6, +19.6] ✓

F THE RESOLUTION-HEADROOM IDENTITY

The identity also holds where it predicts *nothing*. Applying TSR to a crop rather than to the full image gives eight further cells whose headroom is near zero; gain tracks headroom there with $r = 0.911$ and slope 1.03 (mean error 0.40 points), and TSR correctly does nothing. Across both settings the identity is supported on 18 stratum-cells, and it is the only component of this work that predicted a negative result before we ran it.

Two consequences. First, the sign of the gain is predictable before deployment: the one cell where TSR loses (HR-4K cross-instance) is the one cell where the baseline itself is worse at 900 tokens than at 600, so nothing was available to convert. Second, the pruning contributes nothing on its own. Comparing TSR against uniform@900, which encodes at the same resolution and prunes nothing, the two agree within 0.5 points in 7 of the 10 cells (+0.0, +0.0, +0.0, +0.0, +0.2, +0.5, −0.2). In two more the pruned arm is *ahead*, by +1.3 and +2.6, which we read as noise rather than a benefit of pruning. Exactly one cell shows a real cost: HR-8K cross-instance at −1.8 [−3.2, −0.5], the only measurable price of discarding 90% of visual tokens anywhere in this work. Pruning at the boundary is therefore free everywhere we have measured it but that one cell. We verified the pruning is applied rather than silently skipped at the distribution level: output probabilities change on 99% of items while the arg-max changes on 2%. The discarded tokens are inert, not absent.

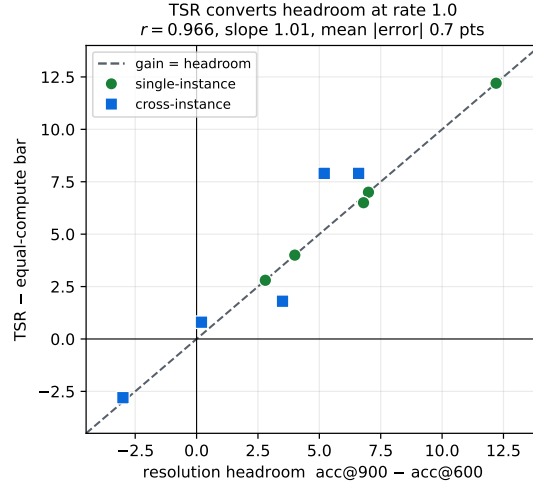


Figure 7: **TSR is a resolution converter.** Its gain over the equal-compute bar tracks the headroom the checkpoint already has, at rate 1.0. The one cell where TSR loses is the one cell where the baseline itself is worse at 900 tokens than at 600.

Table 9: TSR’s gain against the headroom $\text{acc}(900) - \text{acc}(600)$, for the ten stratum-cells that carry the `uniform@900` diagnostic. Pearson $r = 0.966$, OLS slope 1.01, mean $|\text{gain} - \text{headroom}| = 0.7$ points. Headroom is measurable from two baseline runs with no method involved.

cell	stratum	TSR – bar	headroom
V*Bench Qwen2.5-VL-7B	single	+12.2 ✓	+12.2
	cross	+7.9	+5.2
V*Bench Qwen3-VL-8B	single	+7.0	+7.0
	cross	+7.9 ✓	+6.6
HR-4K Qwen3-VL-2B	single	+4.0 ✓	+4.0
	cross	−2.8	−3.0
HR-4K Qwen2-VL-7B	single	+2.8	+2.8
	cross	+0.8	+0.2
HR-8K Qwen3-VL-2B	single	+6.5 ✓	+6.8
	cross	+1.8	+3.5

G SCOPE AND COMPUTE, AS FIGURES

H ADDITIONAL VISUALISATIONS

I DESIGNS THAT DID NOT WORK

The last row generalises the rest, and explains why the two policies cannot be merged into one. TWR’s signal is produced after the transport boundary, and pruning only saves compute at or before it. A pipeline that prunes early enough to save has destroyed the layers TWR reads, which costs 3.1 and 12.6 points of coverage; a pipeline that waits for TWR has nothing left worth pruning. Holding the ranking fixed and moving only the cut from L16 to L4 costs 5.2 points. The incompatibility is structural, not a tuning failure.

The two policies are substitutes, not complements. The structural argument above is not the only reason they do not combine. We also tested the composition that *is* mechanically sound: hold the token-layer budget fixed, and let any placement rule spend its answer pass on a crop encoded at 460 tokens and pruned 90% at the boundary instead of a plain crop at 300. Applied to ViCrop-style block-mean, to LASER, to TWR and to oracle placement, on both checkpoints, every one of the eight contrasts spans zero. The diagnostic says why: the resolution headroom left *on a crop*

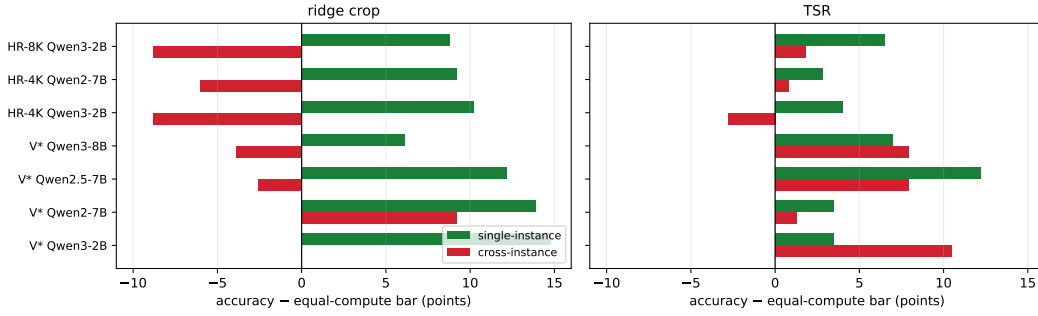


Figure 8: **The scope law.** Cropping is worth +6 to +15 points on single-instance questions in all seven cells and is negative on cross-instance in six of them. TSR is positive on both strata in six of seven, and gives up most of the single-instance gain to get there. The two policies are not variants of one method.

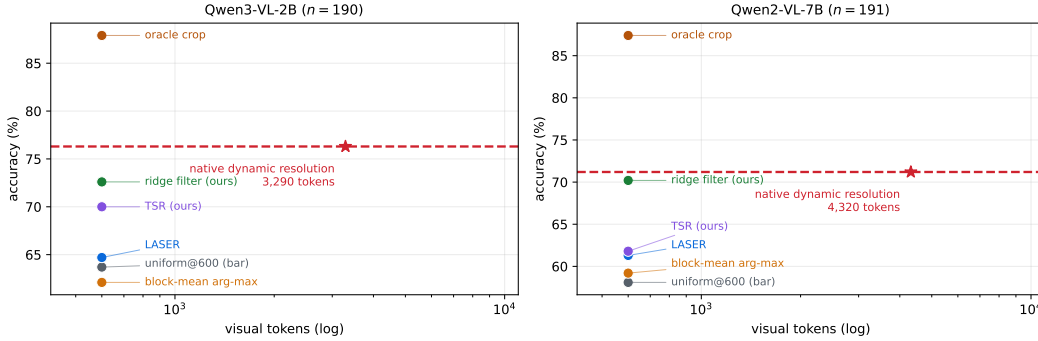


Figure 9: **Accuracy against visual-token compute, V*Bench.** The dashed line is the model run at its own native dynamic resolution. Our two policies sit on it at 14–18% of its compute; the published rules sit significantly below it; and a perfectly placed 600-token crop sits well above it. Placement is worth more than 5–7× the tokens.

averages +0.8 points, against +4.5 on the full image. A 0.25×0.25 window re-encoded at 300 tokens has already lifted the object above the encoding cliff, so the magnification TSR would buy has already been spent. Both policies spend the same resource, effective resolution on the region that matters; cropping spends it in a targeted way and TSR spends it uniformly. That is why their *scope* is complementary while their *gains* are not additive.

Why cropping cannot reach cross-instance questions. Under *perfect* placement, cross-instance accuracy reaches 75.0 and 77.6, an irreducible error of 25% and 22%, against 3.5% and 6.1% for single-instance. Two crops at full resolution covering *both* objects, which raises cross-instance object coverage by 17.1 and 15.8 points, still score -2.6 and -3.9 against a single crop. Cross-instance error is not predominantly perceptual, so no reallocation of perceptual resources reaches it. Resolution does reach part of it, which is what the second policy exploits, but cropping does not.

J PIPELINE FAULTS FOUND, AND HOW THEY WERE CAUGHT

Two runs in this work produced plausible numbers that were wrong. We report them because the rules that caught them are the reason we trust the rest, and because both faults are easy to make and invisible in aggregate accuracy.

A silently unpatched module. Pruning is injected by monkey-patching the attention forward pass. The patch covered two module paths; a third checkpoint’s attention lived in a third module, so pruning was a no-op and the arm was byte-identical to its unpruned control while reporting a clean

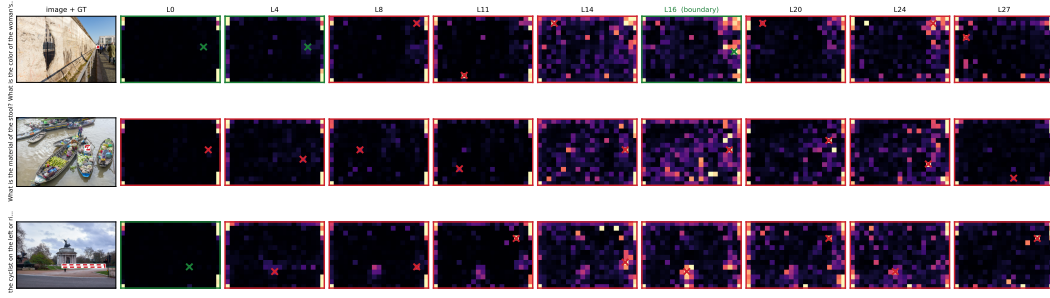


Figure 10: **The same image, the same question, a different answer about where to look at every depth.** Each panel is one layer’s attention over image cells (colour clipped at the 98th percentile so the serialisation sink does not wash out the structure); the cross is that layer’s ring-masked arg-max, green where the resulting crop would cover the evidence and red where it would not. Nothing about the input changes across a row. Only the read-out depth does.

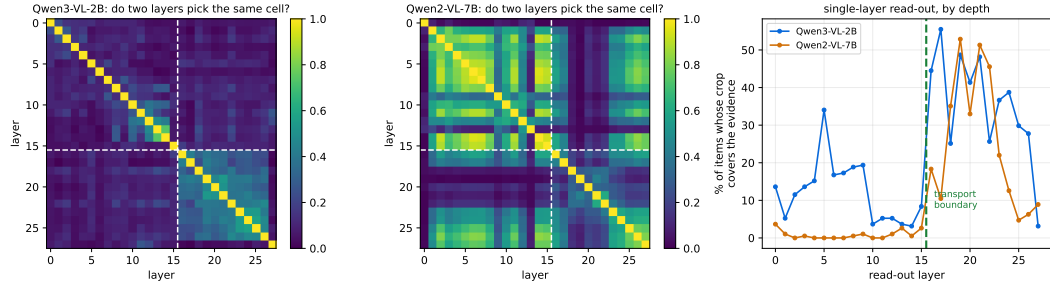


Figure 11: **Left, centre: two layers rarely choose the same cell.** Mean off-diagonal agreement is 13.8% and 38.1%; the block structure splits at the transport boundary (dashed). **Right: and only the layers after the boundary are worth reading.** The fraction of items whose single-layer crop covers the evidence is at or below 20% everywhere before the boundary, jumps immediately after it, and decays again toward the output. This is the paper’s central claim in one curve: the usable placement signal exists only after transport has finished.

+12.2. It was caught by a standing rule: a contrast of exactly +0.0 with a $[+0.0, +0.0]$ interval is a pipeline fault, never a result, because real effects do not land on zero to machine precision. The fix asserts at load time that the model’s actual attention module is one the patch reached, and prints it every run.

A hard-coded depth on a differently-shaped model. The arm table hard-coded the prune layer at 16. On the 36-layer checkpoint the correct depth, $0.57 \times 36 = 21$, was computed and printed but never used, so the run pruned at 44% of depth, inside the transport window. It was caught by the compute budget reading 78% of the bar instead of the expected 98%. We retain the voided run as accidental evidence for the boundary itself: pruning inside the window cost -7.9 pooled and -10.5 cross-instance on a third model family.

Every arm therefore prints its realised token count and its token-layer budget as a fraction of the bar, and any run outside 96–104% is discarded rather than reported.



Figure 12: **Five placement rules, same items, same budget.** Selection rule: single-instance items on which the oracle crop is correct, so the ceiling behaves like a ceiling, and on which TWR is correct while the fixed-layer rule is not; 61 items qualify and the first four by question id are shown. The fixed mid-layer recipe crops glass, masonry and sky.



Figure 13: **Eight inferences, Qwen3-VL-2B.** Top of each cell: the image with the ground-truth box, the block-mean crop and the TWR crop. Below: the two crops the model actually answers from, with its answer.

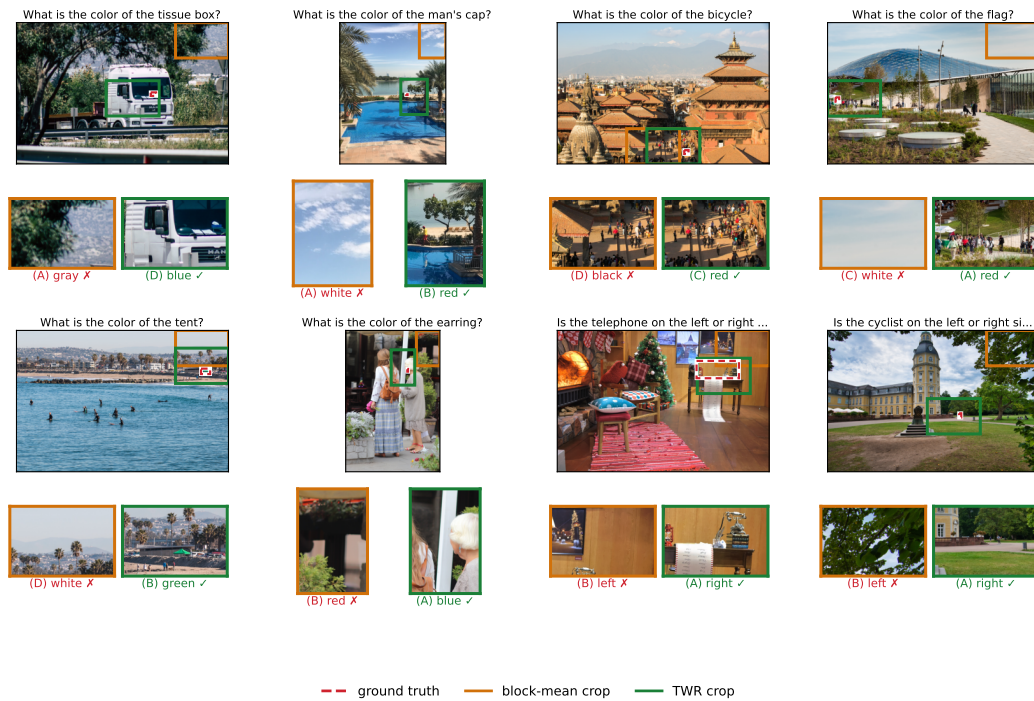


Figure 14: **The same eight-item view on Qwen2-VL-7B.** The failure mode is the same on a different model family and a different parameter count.

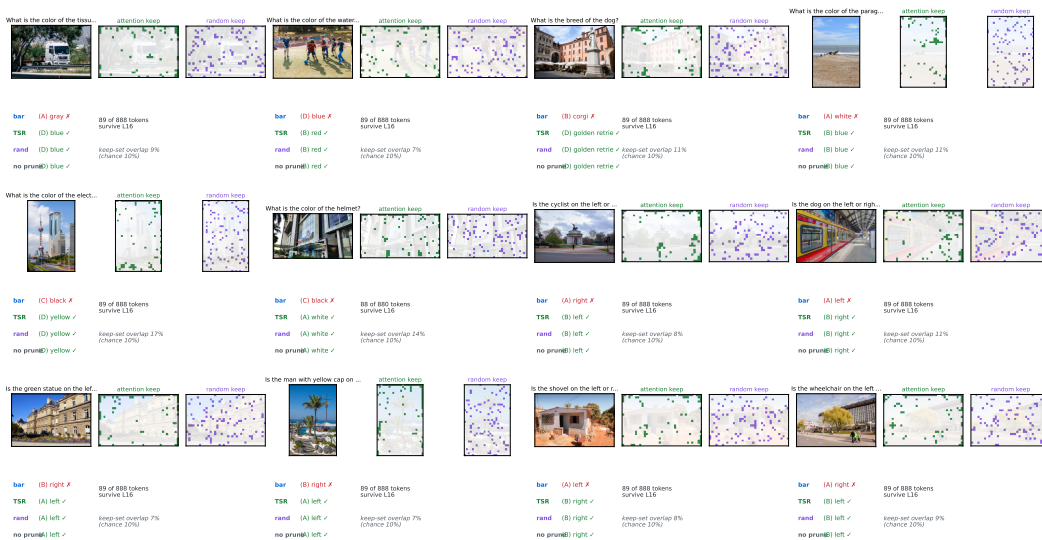


Figure 15: **Twelve inferences under TSR.** For each item: the image, the tokens kept by attention at the boundary, and an equally sized random keep. The two keep-sets overlap at close to the 10% chance rate and the model answers identically under both, and identically again without pruning at all.

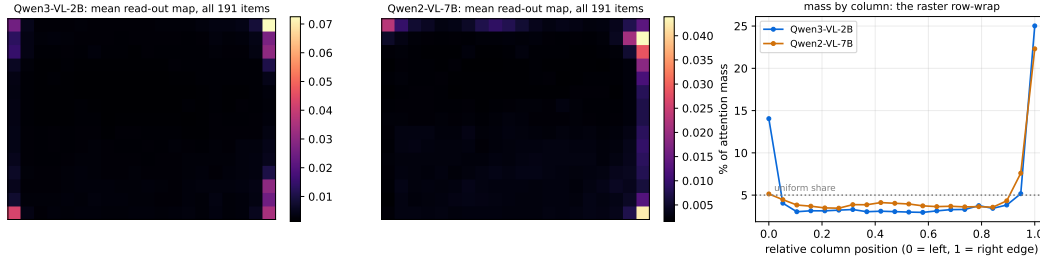


Figure 16: **The serialisation sink, averaged over the benchmark.** Each item’s read-out map is resampled to a common grid and averaged over all 191 items. Mass concentrates in the last column — the token before a raster row-wrap — on both checkpoints, at roughly four times the uniform share. This is what the outer-ring mask in every deployed implementation is patching around.

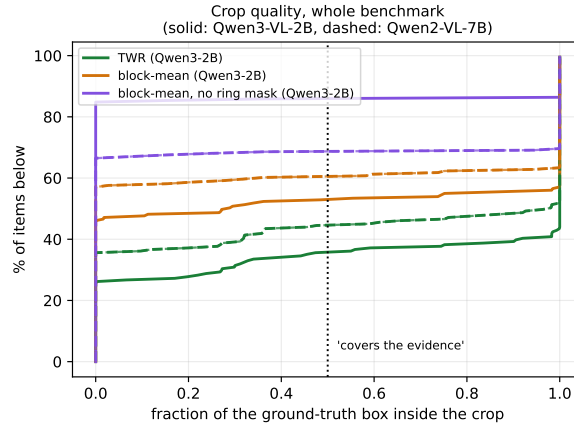


Figure 17: **Crop quality over the whole benchmark.** Cumulative distribution of the fraction of the ground-truth box captured by the crop. Without the ring mask the block-mean rule is close to useless; with it, it is usable; TWR is better than both and barely uses the mask.

Table 10: The ten designs intended to extend cropping to cross-instance questions, and the two that tried to combine the two policies. Each was pre-registered with a primary contrast and a guard before it ran. “Cleared” counts checkpoints of two on which the primary passed.

design	what it changed	cleared	diagnosed cause
per-item window sizer	crop size from map features	0/2	grid inversion, weak features
peak-spanning window	box spanning the top peaks	0/2	relative signal, absolute constant
mass-containment window	window holding $x\%$ of mass	0/2	boxes far too large
joint cell-window ranking	arg-max over both at once	0/2	metric artefact: coverage rises with size
expected-accuracy window	window by predicted accuracy	0/2	accuracy–size curve flat against noise
thumbnail in answer pass	crop plus 64-token scene	0/2	second object below the encoding cliff
global–local split	scene@150 plus crop@150	0/2	halving the crop costs more than the scene gives
context-continued crop	scene@300 plus crop@300	0/2	scene costs 8.7 and 5.2 even with a perfect crop
multi-crop, matched budget	2×150 and 3×100	0/2	magnification tax (-11.3 , -4.3) exceeds coverage gain ($+7.8$, $+2.6$)
funded multi-crop	2×300 , pruned, early exit	0/2	$+3.7$ on one model, -2.6 on the other
TWR-guided early pruning	rank at L4 instead of L16	0/2	-5.2 : ranking before transport is worthless
TSR stacked on a placement rule	crop@460 pruned at L16, same budget	0/2	no resolution headroom left on a crop ($+0.8$ vs $+4.5$)